

Three-Dimensional Finite Element Analysis of the Effect of Angled Abutment Variations (0°, 15°, and 20°) on Stress Distribution in Mandibular First Molar Dental Implants under Axial Loading

Muhammad Farrel Pratama Wicaksono^{1,2}, Muhamad Farel Bimantyas^{1,2}
Dhiaulhauqqe^{1,3}, Kriswanto^{1,2,3}, Tri Indah Winarni^{1,4}, Jamari^{1,2*}

¹Undip Biomechanics Engineering & Research Centre (UBM-ERC), Central Laboratory for Research and Services, Universitas Diponegoro, Semarang 50275, Central Java, Indonesia

²Department of Mechanical Engineering, Faculty of Engineering, Universitas Diponegoro, Semarang 50275, Central Java, Indonesia

³Department of Mechanical Engineering, Faculty of Engineering, Universitas Negeri Semarang, Semarang 50229, Central Java, Indonesia

⁴Department of Anatomy, Faculty of Medicine, Universitas Diponegoro, Semarang 50275, Central Java, Indonesia, UK

*E-mail: j.jamari@gmail.com

Abstract. This in silico study evaluates the biomechanical effects of angled abutments (0°, 15°, and 20°) on a vertical dental implant (90°) in the mandibular first molar region under axial loading. A 3D finite element model was reconstructed from patient CT-scan data using SolidWorks 2023 and analyzed via ANSYS 2022 R1. Cortical and cancellous bones were modeled as linear-elastic orthotropic materials, assuming perfect osseointegration. A static vertical load of 100 N was applied to the occlusal surface. The numerical results revealed that the highest von Mises stress concentration was consistently localized at the implant neck in contact with the cortical crestal bone. Increasing the abutment angle from 0° to 15° and 20° induced greater force eccentricity, generating a bending moment that progressively elevated the peak stress and strain in the surrounding cortical bone. In conclusion, higher abutment angulation significantly increases localized mechanical stress at the bone-implant interface. The 0° straight abutment demonstrates superior biomechanical stability, minimizing the risk of marginal crestal bone loss in the posterior mandible.

1. Introduction

Dental implants have become a highly predictable and widely accepted treatment modality for the rehabilitation of completely and partially edentulous patients. In the posterior region, specifically the mandibular first molar, implants are subjected to significant masticatory forces[1], [2]. Ideally, implants are positioned perpendicularly to the occlusal plane to ensure that the masticatory loads are distributed evenly along the long axis of the implant body. However, anatomical limitations such as severe alveolar bone resorption, the proximity to the inferior alveolar nerve, or adjacent tooth roots often preclude this ideal vertical placement[3], [4]. In such clinical scenarios, angled

abutments are frequently utilized to restore proper occlusal function, alignment, and aesthetics.

While angled abutments offer an essential restorative solution, they fundamentally alter the biomechanical load transmission at the bone-implant interface. The angulation inherently introduces force eccentricity, which generates bending moments even under purely axial occlusal loading. This eccentric loading mechanism can significantly alter localized mechanical stress and strain, particularly around the crestal cortical bone region. Over time, poorly managed stress concentrations in this area can lead to marginal crestal bone loss, risking catastrophic mechanical failures of the implant system[5], [6].

In the realm of modern dental clinical practice, unexpected mechanical failures and long-term biological complications do not only affect patient outcomes but also pose significant occupational health hazards to dental practitioners. Unpredictable implant complications often lead to high-stakes revision procedures, which substantially elevate the cognitive workload, mental stress, and clinical anxiety of clinicians. Furthermore, performing complex, unplanned corrective surgeries increases the duration of awkward static postures, exacerbating the risk of work-related musculoskeletal disorders (WMSDs) among dentists. Therefore, establishing evidence-based clinical protocols and occupational standards rooted in predictable mechanical data is vital to safeguard practitioner well-being and optimize workplace safety in dental clinics.

Finite Element Analysis (FEA) has been extensively validated as a reliable, non-invasive computational method to simulate and evaluate complex biomechanical behaviors in dental implantology. By reconstructing anatomically accurate three-dimensional models from patient CT-scan data, FEA allows researchers to precisely quantify internal stress distributions that are impossible to measure directly *in vivo*[7], [8]. Utilizing FEA data enables the pre-emptive assessment of mechanical risks before clinical implementation.

Accordingly, the primary objective of this study is to evaluate and compare the biomechanical effects of varying abutment angulations (0°, 15°, and 20°) on a vertically positioned dental implant in the mandibular first molar region under pure axial loading. By quantifying the resulting von Mises stress and strain distributions via high-fidelity 3D computational simulation, this study aims to provide a rigorous mechanical baseline. Ultimately, these findings serve as a foundation for developing standardized clinical guidelines and occupational health protocols, thereby reducing procedural uncertainty, minimizing occupational stress, and enhancing ergonomic safety for dental professionals during restorative interventions.

2. Material and Methods

2.1 Geometric Modeling and Reconstruction

The three-dimensional (3D) geometric model of the mandibular first molar region was reconstructed using patient computed tomography (CT) scan data to ensure anatomical accuracy. The solid model was processed and assembled using computer-aided design (CAD) software, SolidWorks 2023. The total assembly comprised the dental implant fixture, the bone block (differentiated into cortical and cancellous bone structures), and the abutment component. To systematically evaluate the mechanical impact of geometric variations on clinical safety, three distinct configurations were established based on the abutment angulation: 0° (straight), 15°, and 20° relative to the long axis of the vertical implant body.

2.2 Material Properties

In this *in silico* simulation, all components within the dental implant-bone system were treated as linear-elastic. To accurately replicate the biological nature of human bone, the cortical and cancellous bone tissues were modeled as orthotropic materials, exhibiting direction-dependent mechanical properties. Conversely, the dental implant and abutment assembly were modeled as a homogeneous, isotropic titanium alloy (Ti-6Al-4V), a standard material widely utilized in clinical practice due to its high biocompatibility and mechanical strength. The specific mechanical properties used in this simulation including Young's modulus and Poisson's ratio are summarized in Table 1[9].

Table 1. Mechanical properties of the bone tissues and implant components.

Bagian	Modulus Young (GPa)	Shear Modulus (GPa)	Possion's Ratio	Density
Cortical Bone	$E_x = 12,7$	$G_{xy} = 5.5$	$\nu_{xy} = 0.31$	-
	$E_y = 22.8$	$G_{yz} = 7.4$	$\nu_{yz} = 0.28$	-
	$E_z = 17.9$	$G_{xz} = 5.0$	$\nu_{xz} = 0.18$	-
Cancellous Bone	$E_x = 0.511$	$G_{xy} = 0.434$	$\nu_{xy} = 0.31$	-
	$E_y = 0.907$	$G_{yz} = 0.081$	$\nu_{yz} = 0.30$	-
	$E_z = 0.114$	$G_{xz} = 0.078$	$\nu_{xz} = 0.22$	-
Implant Complex	113.8		0.342	4.43

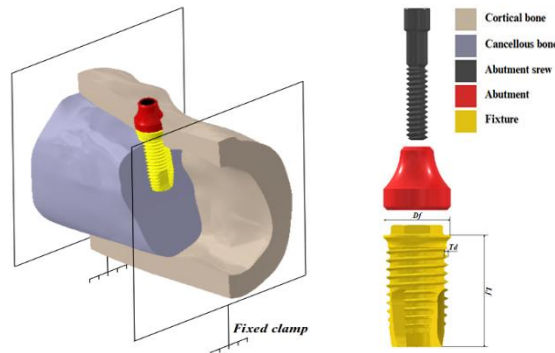


Figure 1 3D Model Posterior Mandibula First Molar Implant

2.3 Finite Element Meshing and Interface Conditions

The complete CAD assemblies were imported into ANSYS 2022 R1 for numerical discretization and finite element analysis. High-order 3D solid elements (quadratic tetrahedral elements) were implemented to mesh the complex geometries of both the bone structures and the implant threads. A mesh refinement protocol was conducted around the implant-bone interface to capture localized stress gradients accurately. To simulate an ideal, fully healed clinical outcome (perfect osseointegration), the interface condition between the outer surface of the implant threads and the surrounding bone surfaces was designated as a "bonded" contact type. This configuration prevents any relative sliding or separation at the interface, ensuring continuous load transmission.

2.4 Boundary Conditions and Loading Configurations

To simulate realistic physiological constraints within the mandibular architecture, fixed boundary conditions (zero degrees of freedom for both displacement and rotation) were applied to the base and mesio-distal peripheral surfaces of the bone block. For the loading configuration, a static, purely vertical (axial) load of 100 N was applied uniformly to the top occlusal surface of the abutment. From an occupational health and clinical standardisation perspective, utilizing a standardized, static 100 N axial load isolates the independent variable of abutment angulation. Establishing this predictable, controlled loading baseline allows for the development of reproducible clinical guidelines. This mechanical predictability is vital for reducing the cognitive workload and procedural uncertainty of dental practitioners, ultimately mitigating workplace stress and ergonomic hazards during complex prosthodontic seating procedures.

3. Result and Discussion

3.1 Numerical Results and Stress Distribution

The finite element simulations successfully quantified the equivalent (von Mises) stress distributions across the dental implant complex under a static vertical load of 100 N. The numerical analysis revealed that the maximum von Mises stress was consistently localized at the implant neck, specifically at the first micro-thread in direct contact with the crestal cortical bone layer. Interestingly, the computational data demonstrated a progressive decrease in peak mechanical stress as the abutment angulation increased. For Model B (15° abutment angulation), the maximum equivalent stress was recorded at 30.48 MPa. In Model C (20° abutment angulation), the peak stress decreased to 26.268 MPa. Finally, for Model D (25° abutment angulation), the maximum stress further dropped to $2.2797 \times 10^7 Pa$, which is equivalent to 22.797 MPa. The consolidated maximum stress values for each model are presented in Table 2, following the formal academic formatting guidelines.

Table 2. Maximum von Mises stress values across different abutment angulations.

Model Configuration	Abutment Angulation	Maximum von Mises Stress (MPa)	Critical Stress Location
Model A	15°	30.480	Implant Neck
Model B	20°	26.268	Implant Neck
Model C	25°	22.797	Implant Neck

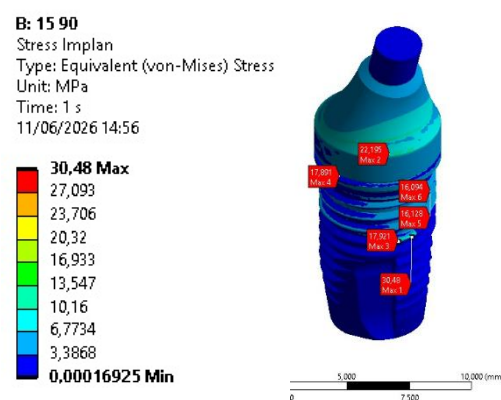


Figure 1. Stress Distribution for Implant 19°

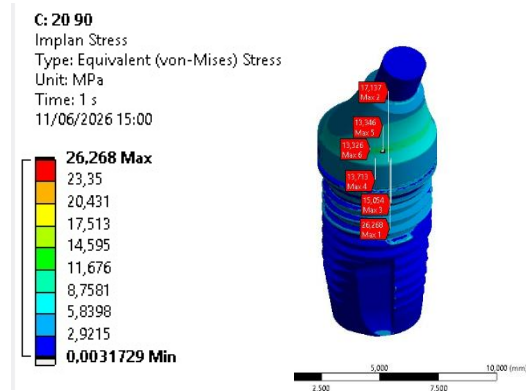


Figure 3. Stress Distribution for Implant 20°

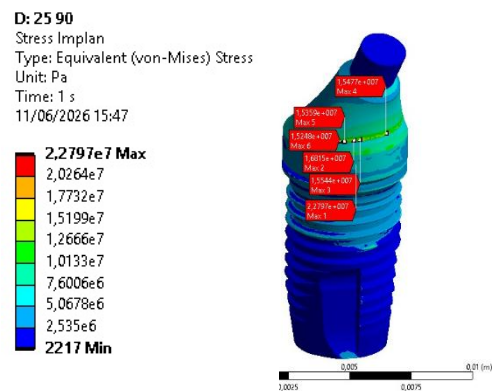


Figure 4. Stress Distribution for Implant 25°

3.2 Biomechanical Behavior and Stress Redistribution

The localization of the maximum von Mises stress at the implant neck across all configurations confirms that the bone-implant interface at the crestal cortical level acts as the primary fulcrum for load transmission. Cortical bone, possessing a significantly higher Young's modulus than cancellous bone, naturally bears the brunt of the mechanical forces[10], [9].

The counterintuitive descending trend of peak stress from 15° to 25° under a purely vertical load can be attributed to the geometric alteration of the load transfer path. Under a constant vertical force of 100 N, increasing the abutment angulation shifts the loading point laterally relative to the implant's long axis. At a lower angulation (15°), the force vector creates a highly concentrated compressive stress state at the crestal micro-threads. As the angulation increases to 20° and 25°, the geometric reorientation introduces a larger bending moment component that alters the internal stress state from highly localized compression to a combined tension-compression state. This phenomenon partially redistributes the stress fields across a broader area of the implant geometry, thereby lowering the absolute peak von Mises stress at that specific localized hotspot[11].

3.3 Implications for Occupational Health Standards in Dental Practice

Integrating these computational biomechanical findings into clinical workflows yields vital implications for establishing occupational health and safety standards within dental environments. In conventional dental clinical practice, structural failures of implants—such as thread stripping, screw loosening, or marginal bone loss—represent significant sources of occupational stress and cognitive burnout for dental practitioners[12], [13]. When an implant fails catastrophically *in vivo*, the clinician is forced to perform unplanned, high-risk revision surgeries. These corrective procedures are highly demanding, significantly elevating the dentist's mental fatigue, anxiety, and perceived workload[14]. From an ergonomic standpoint, complex revision surgeries require dentists to maintain prolonged, awkward static postures, directly compounding the risk of work-related musculoskeletal disorders (WMSDs). By providing clear, predictable mechanical baselines—such as demonstrating that angled abutments between 15° and 25° exhibit stable, predictable stress redistribution under vertical loads—this study contributes to evidence-based clinical protocols. Standardizing implant selection and installation guidelines based on FEA data minimizes the incidence of mechanical failures. Consequently, reducing procedural uncertainty directly mitigates the clinical anxiety of dental professionals, lowers their psychological stress, and protects them from the physical hazards of prolonged corrective

interventions, fostering a safer and more sustainable workplace environment.

4. Conclusion

This *in silico* computational study successfully evaluated the three-dimensional stress distribution within mandibular first molar dental implants utilizing variable angled abutments under static vertical loading. The numerical results from the finite element analysis demonstrated that the maximum equivalent von Mises stress is consistently concentrated at the implant neck interface adjacently contacting the crestal cortical bone. Under a standardized pure axial load of 100 N, increasing the abutment angulation from 15°, 20°, 25° yielded a progressive reduction in peak mechanical stress from 30.480 MPa, 26.268 MPa, to 22.797 MPa, respectively, owing to favorable geometric stress redistribution along the restorative assembly.

From an occupational health perspective, these findings provide a highly predictable mechanical baseline that is essential for the formulation of standardized dental clinical protocols. By establishing clear biomechanical guidelines regarding the load-bearing capabilities of angled abutments, dental practitioners can substantially minimize the risk of post-surgical mechanical failures. Consequently, this risk reduction directly mitigates the clinical anxiety, cognitive overloading, and ergonomic strains heavily associated with complex, extended revision surgeries, thereby fostering a safer, more sustainable, and ergonomically sound workplace environment for dental professionals.

5. References

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